

1. Yes, magnesium and oxygen form an ionic compound with the empirical formula MgO.
2. 8.4 mg
3. H_3PO_4
4. $6.4 \times 10^0 \text{ mm}$
5. Systematic error $\therefore$ every measurement is consistently shifted by the same amount in the same direction.
6. $558.7 \frac{\text{kg}}{\text{m}^3}$
7. 0.0045 m
8. 31.6 cm
9. Pure substances: A, C $\therefore$ the mass ratio of E_1 to E_2 in A and C is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
10. $0.24 \frac{\text{g}}{\text{mL}}$
11. Ga
12. $7.77 \times 10^{11} \text{ Hz}$
13. $4.289 \times 10^0 \text{ g}$
14. Homogeneous mixture $\therefore$ it is a combination of two or more substances with a uniform composition and appearance.
15. 140°F
16. 4.8% $\therefore$ $\% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
17. Random error $\therefore$ the measurements fluctuate unpredictably in both directions.
18. $1.3 \times 10^{-3} \text{ m}$
19. 2 protons and 2 neutrons.
20. 16 mL
21. Liquid
22. It is a homogeneous mixture. $\therefore$ it is a combination of two or more substances with a uniform composition and appearance.
23. Mn
24. 2.641 kg
25. The SI base unit for temperature is the kelvin (symbol: K).
26. Compound
27. 0.000000058 m
28. $8.55 \times 10^1 \text{ fm}$
29. HClO_2
30. 0.000000079 m
31. 16.767 g
32. Mixture. It is a (likely heterogeneous) mixture; it contains minerals, organic matter, air, and water in varying proportions.
33. It is an element. $\therefore$ it contains only one element and is pure in its composition.
34. HCl
35. -30°C
36. $2. \times 10^3 \text{ mol}$
37. X is a pure substance; Mass of E_1 : 6.4 g. X is a pure substance $\therefore$ the ratio of the masses of E_1 to E_2 is the same for all four samples. A pure substance always has a constant composition. If $m_{\text{E}_1} + m_{\text{E}_2} = 10.0 \text{ g}$, and you have calculated the mass fraction $m_{\text{E}_1}/(m_{\text{E}_1} + m_{\text{E}_2})$ from the samples, you would expect m_{E_1} to be 6.4 g.
38. $7.70 \times 10^{-1} \text{ m}$
39. $3.45 \frac{\text{g}}{\text{mL}}$
40. $6.12 \times 10^{-1} \text{ m}$

41. SCl_2
42. Barium thiocyanate: $\text{Ba}(\text{SCN})_2$
43. Systematic error $\because$ every measurement is consistently too high by the same amount due to a calibration flaw.
44. 0.000021 m
45. No, chlorine and fluorine cannot form an ionic compound.
46. 19.52 mL
47. Compound $\because$ it contains carbon and hydrogen atoms bonded in a fixed ratio.
48. dibromine heptoxide
49. 1.5 mm
50. $9 \frac{\text{mol}}{\text{L}}$
51. Sample B $\because \rho_A = 0.42 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 0.69 \frac{\text{g}}{\text{mL}}$
52. $9.29 \times 10^1 \text{ fm}$
53. 0.0058 m
54. 58.4 fm
55. 18 protons and 22 neutrons.
56. $6.4 \times 10^0 \text{ mg}$
57. 3.3% $\because \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
58. 20.241 g
59. $20 \frac{\text{mol}}{\text{L}}$
60. $3.4 \times 10^{-11} \text{ g}$
61. $9.804 \times 10^1 \text{ MPa}$
62. Si is a neutral atom. It has 14 protons and 14 electrons.
63. 122 °F
64. 79.3 Gm
65. 31 g
66. $9.3 \times 10^{-14} \text{ s}$
67. $3.7 \times 10^1 \text{ g}$
68. Compound
69. No new substance is formed $\because$ Sodium and chloride ions simply disperse through the water.
70. Be
71. Krypton
72. $8.6645 \times 10^2 \text{ nm}$
73. 83.7 Gm
74. Barium oxide tetrahydrate: $\text{BaO} \cdot 4\text{H}_2\text{O}$
75. Iron (III) oxalate: $\text{Fe}_2(\text{C}_2\text{O}_4)_3$
76. 6.631 mL
77. 280 mL
78. CO_2
79. 5836.4 g
80. dibromine monoxide
81. $5.0 \times 10^{-1} \text{ m}$
82. 293 K
83. The SI base unit for time is the second (symbol: s).
84. Homogeneous mixture $\because$ it is a combination of two or more substances with a uniform composition and appearance.

85. 32 mL
86. Iron most commonly forms the cation Fe^{3+} .
87. 24.2 cm
88. $1 \text{ kg} = 10^3 \text{ g}$
89. 281000000 W
90. Chemical property. The reactivity with acid of zinc is a chemical property; it describes zinc's tendency to react with acids to produce new substances.
91. $81.9 \mu\text{g}$
92. 6.767 kg
93. Yes, potassium and chlorine form an ionic compound with the empirical formula KCl.
94. 4.9 mg
95. $8.21 \times 10^8 \text{ W}$
96. Element
97. $0.329 \frac{\text{g}}{\text{mL}}$
98. 900000000000 Hz
99. $2.72 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
100. 76.95 MPa
101. 6.474 g
102. 20°C
103. 232000000000 Hz
104. Caesium fluoride: CsF
105. Gas
106. $5.8 \times 10^{-11} \text{ g}$
107. $9.0 \times 10^1 \text{ g}$
108. Heterogeneous mixture $\therefore$ it is a combination of two or more distinct substances that are not uniformly distributed.
109. $9.3100 \times 10^{-3} \text{ mL}$
110. $4.527 \times 10^0 \text{ km}$
111. Br_4Si is a molecular compound.
112. The SI base unit for time is the second (symbol: s).
113. 273000000 W
114. X is a pure substance; Mass of E_1 : 6.9 g. X is a pure substance $\therefore$ the ratio of the masses of E_1 to E_2 is the same for all four samples. A pure substance always has a constant composition. If $m_{E1} + m_{E2} = 10.0 \text{ g}$, and you have calculated the mass fraction $m_{E1}/(m_{E1} + m_{E2})$ from the samples, you would expect m_{E1} to be 6.9 g.
115. Physical change $\therefore$ water changes from liquid to vapor but its chemical identity remains water.
116. Pure substance. It is a pure substance (element); diamond consists entirely of carbon atoms in a crystal lattice.
117. Zinc fluoride tetrahydrate: $\text{ZnF}_2 \cdot 4\text{H}_2\text{O}$
118. 4.8% $\therefore \% \text{ Error} = \left| \frac{E-A}{A} \right| \times 100\%$
119. $3.3 \frac{\text{m}}{\text{s}}$
120. $42.8 \mu\text{g}$
121. $1.0 \times 10^{-1} \text{ m}$
122. Silver chromate tetrahydrate: $\text{Ag}_2\text{CrO}_4 \cdot 4\text{H}_2\text{O}$
123. Physical change $\therefore$ the sugar molecules disperse into the water but remain sugar molecules.
124. 0.51 m
125. 303 K
126. 5178.6 m

127. Boric acid
128. 0.675 m
129. $6.0 \times 10^{-14} \text{ s}$
130. $7.83 \times 10^1 \text{ Gm}$
131. $2.92 \times 10^1 \text{ cm}$
132. $1 \text{ Mg} = 10^6 \text{ g}$
133. Carbonic acid
134. $20 \frac{\text{mol}}{\text{L}}$
135. Chlorous acid
136. 0.000000044 m
137. Extensive property. Weight is an extensive property $\therefore$ the total weight depends on how much matter is present.
138. $4.340 \times 10^0 \text{ g}$
139. 519.84 nm
140. 268.8 K
141. Element $\therefore$ it contains only iron atoms.
142. $3.553 \times 10^3 \frac{\text{kg}}{\text{m}^3}$
143. Zn^{2+} is a cation. It has 30 protons and 28 electrons.
144. Tin
145. 3.6 mm
146. Calcium chlorite: $\text{Ca}(\text{ClO}_2)_2$
147. $10.8 \frac{\text{m}}{\text{s}}$
148. Pure substance. It is a pure substance (compound); it has a definite molecular formula and a fixed composition.
149. 5.347 mL
150. Chemical property. The reactivity with acid of zinc is a chemical property; it describes zinc's tendency to react with acids to produce new substances.
151. 220 g
152. 4285.2 g
153. Intensive property. Density is an intensive property $\therefore$ the ratio of mass to volume does not depend on how much of the substance is present.
154. 0.000088 m
155. $7.065 \times 10^1 \text{ MPa}$
156. Calcium chloride: CaCl_2
157. Silver fluoride: AgF
158. Pure substances: A $\therefore$ the mass ratio of E_1 to E_2 in A is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
159. 288 K
160. Sample B $\therefore \rho_A = 3.3 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 10. \frac{\text{g}}{\text{mL}}$
161. Chemical change $\therefore$ the chemicals react to produce carbon dioxide gas.
162. Chemical property. The toxicity of carbon monoxide is a chemical property; it describes carbon monoxide's ability to react with hemoglobin, forming a new compound.
163. 360 K
164. Extensive property. Mass is an extensive property $\therefore$ the total mass depends on how much substance is present.
165. 0.000000000072 g
166. Element $\therefore$ it contains only gold atoms.
167. Sample A $\therefore \rho_A = 2.4 \frac{\text{g}}{\text{mL}}$ and $\rho_B = 1.5 \frac{\text{g}}{\text{mL}}$

168. No new substance is formed $\therefore$ The wire changes shape but remains copper.
169. $-35\text{ }^{\circ}\text{C}$
170. $3.7\frac{\text{g}}{\text{mL}}$
171. $3.42\frac{\text{m}}{\text{s}}$
172. 6.186 kg
173. $3.249 \times 10^0\text{ km}$
174. $10.4\frac{\text{g}}{\text{mL}}$
175. $2.3 \times 10^{-5}\text{ m}$
176. $1.15 \times 10^2\text{ g}$
177. Plutonium
178. 27 protons and 32 neutrons.
179. $(\text{H}_3\text{O})_3\text{PO}_4$ is an ionic compound.
180. $1.6806 \times 10^3\text{ m}$
181. $-4\text{ }^{\circ}\text{F}$
182. Gas
183. carbon dioxide
184. It is a compound. $\therefore$ it is a pure substance formed when two or more different chemical elements bond together in a fixed ratio.
185. 343 K
186. $\text{Cu}(\text{OH})_2$ is an ionic compound.
187. Lithium most commonly forms the cation Li^+ .
188. 900 mol
189. Silver most commonly forms the cation Ag^+ .
190. $1.8 \times 10^3\text{ mol}$
191. Ag is a neutral atom. It has 47 protons and 47 electrons.
192. Yes, a new substance is formed $\therefore$ Water is produced from hydrogen and oxygen gas.
193. $1.1\frac{\text{g}}{\text{mL}}$
194. 7683.1 m
195. Pure substances: B, C $\therefore$ the mass ratio of E_1 to E_2 in B and C is approximately an integer multiple of the ratio in pure substance X. When two elements form different pure compounds, the masses of one element that combine with a fixed mass of the other are related by integer ratios (Law of Multiple Proportions).
196. $5.7755 \times 10^3\text{ g}$
197. N_2O
198. $0.000000000000017\text{ s}$
199. 10^3
200. $66.3\text{ }\mu\text{g}$